Solutions for
Analysis I&II (Forth Edition)
by Terence Tao

Ilia Koloiarov

June 30, 2026

Contents

Chapter 1

Introduction

No exercises in this chapter.

Chapter 2

Starting at the beginning: the natural numbers

2.1 The Peano axioms

Axiom 2.1. 0 *is a natural number.*

Axiom 2.2. *If* n *is a natural number, then* $n{+}{+}$ *is also a natural number.*

Axiom 2.3. 0 *is not the successor of any natural number; i.e., we have* $n{+}{+} \neq 0$ *for every natural number* n*.*

Axiom 2.4. *Different natural numbers must have different successors; i.e., if* n, m *are natural numbers and* $n \neq m$*, then* $n{+}{+} \neq m{+}{+}$*. Equivalently, if* $n{+}{+} = m{+}{+}$*, then we must have* $n = m$*.*

Axiom 2.5. (Principle of mathematical induction). *Let* $P(n)$ *be any property pertaining to a natural number* n*. Suppose that* $P(0)$ *is true, and suppose that whenever* $P(n)$ *is true,* $P(n{+}{+})$ *is also true. Then* $P(n)$ *is true for every natural number* n*.*

Definition 2.1.3. *We define* 1 *to be the number* $0{+}{+}$*,* 2 *to be the number* $(0{+}{+}){+}{+}$*, etc.*

Proposition 2.1.4. 3 *is a natural number.*

Proof. Given the Axiom 2.1, 0 is a natural number. $1 := 0{+}{+}$ is a natural number by Axiom 2.2. Same as $2 := 1{+}{+}$ and $3 := 2{+}{+}$. □

Proposition 2.1.6. 4 *is not equal to* 0.

Proof. By Definition 2.1.3, $4 := 3{+}{+}$, thus $3{+}{+} \overset{\lightning}{=} 0$ what contradicts Axiom 2.3 □

Proposition 2.1.8. 6 *is not equal to* 2

Proof. Let's assume that $6 = 2$. By Definition 2.1.3, $6 = 5{+}{+}$ and $2 = 1{+}{+}$. That means, $5 = 1$ by Axiom 2.4. Repeating the procedure, we end up with $4 \overset{\lightning}{=} 0$ what contradicts previously proven Proposition 2.1.6 □

Lemma 2.1.8.a. *For any natural number n, $n \neq n{+}{+}$.*

Proof. For the base case, $0 \neq 0{+}{+}$ by Axiom 2.3. For the step case, we need to prove if $n \neq n{+}{+}$, then $n{+}{+} \neq (n{+}{+}){+}{+}$. Let's assume that $n{+}{+} = (n{+}{+}){+}{+}$. Then by Axiom 2.4, $n \overset{\lightning}{=} n{+}{+}$, what contradicts the inductive hypothesis. Now we have closed the induction. □

Proposition 2.1.16. *(Recursive definitions). Suppose for each natural number n, we have some function $f_n : \mathbb{N} \to \mathbb{N}$ from the natural numbers to the natural numbers. Let c be a natural number. Then we can assign a unique natural number to each natural number n, such that $a_0 = c$ and $a_{n++} = f_n(a_n)$ for each natural number n.*

Proof. First, none of a_n redefines a_0 by Axiom 2.3. Second, let's assume that the procedure gives the unique value for a_n. Then it also provides the unique value for a_{n++} by Axiom 2.4 what closes the induction. □

2.2 Addition

Definition 2.2.1. *(Addition of natural numbers). Let m be a natural number. To add zero to m, we define $0+m := m$. Now suppose inductively that we have defined how to add n to m. Then we can add $n{+}{+}$ to m by defining $(n{+}{+}) + m := (n+m){+}{+}$.*

Proposition 2.2.1.a. *The sum of two natural numbers is again a natural number.*

Proof. We induct on n. Using Definition 2.2.1, $0 + m := m$ is a natural number by the proposition's assumption. Let's assume that $n + m$ is a natural number. Then $(n{+}{+}) + m := (n + m){+}{+}$ by Definition 2.2.1 and $(n + m){+}{+}$ is a natural number by Axiom 2.2. That closes the induction. □

Lemma 2.2.2. *For any natural number n, $n + 0 = n$.*

Proof. Guess what, we use induction. Given that 0 is a natural number and Definition 2.2.1, $0 + 0 := 0$ what proves the base case. Let's assume that $n + 0 := n$, then $(n{+}{+}) + 0 := (n + 0){+}{+}$ using Definition 2.2.1, and $(n + 0){+}{+} = n{+}{+}$ by the induction assumption what proves the step case. That closes the induction. □

Lemma 2.2.3. *For any natural numbers n and m, $n + (m{+}{+}) = (n + m){+}{+}$.*

Proof. Let's induct on n. To prove the base case, $0+(m{+}{+}) := m{+}{+} := (0+m){+}{+}$ using Definition 2.2.1. For the step case, we need to prove $(n{+}{+})+(m{+}{+}) = ((n{+}{+})+m){+}{+}$. Let's assume $n + (m{+}{+}) = (n + m){+}{+}$, then

$$\begin{aligned}
(n{+}{+}) + (m{+}{+}) &:= (n + (m{+}{+})){+}{+} && \text{Definition 2.2.1}\\
&= ((n + m){+}{+}){+}{+} && \text{Inductive Hypothesis}\\
&= ((n{+}{+}) + m){+}{+} && \text{Definition 2.2.1}
\end{aligned}$$

what closes the induction. □

Corollary 2.2.3.a. *For any natural number n, $n{+}{+} = n + 1$.*

Proof. The base case is set up by Definition 2.1.3. For the step case, we need to prove $(n{+}{+}){+}{+} = (n{+}{+}) + 1$, assuming $n{+}{+} = n + 1$:

$$\begin{aligned}
(n{+}{+}){+}{+} &= (n + 1){+}{+} && \text{Inductive Hypothesis}\\
&= (n{+}{+}) + 1 && \text{Definition 2.2.1}
\end{aligned}$$

□

Proposition 2.2.4. (Addition is commutative). *For any natural numbers n and m, $n + m = m + n$.*

Proof. Let's (again) induct on n. For the base case we need to prove $0 + m = m + 0$. The left term equals to m by Definition 2.2.1. The right term equals also to m by Lemma 2.2.2. Thus, both terms are equal. For the step case, we need to prove $(n{+}{+}) + m = m + (n{+}{+})$, assuming $n + m = m + n$:

$$\begin{aligned}
(n{+}{+}) + m &= (n + m){+}{+} && \text{Definition 2.2.1} \\
&= (m + n){+}{+} && \text{Inductive Hypothesis} \\
&= (m + (n{+}{+})) && \text{Lemma 2.2.3}
\end{aligned}$$

what closes the induction. □

Proposition 2.2.5. (Addition is associative). *For any natural numbers a, b, c, we have $(a + b) + c = a + (b + c)$.*

Proof. Let's induct on a. For the base case, we need to prove

$$(0 + b) + c = 0 + (b + c)$$

$$\text{Definition 2.2.1} \qquad b + c = b + c \qquad \text{Lemma 2.2.2}$$

For the step case, we need to prove $((a{+}{+}) + b) + c = (a{+}{+}) + (b + c)$, assuming $(a + b) + c = a + (b + c)$:

$$\begin{aligned}
((a{+}{+}) + b) + c &= ((a + b){+}{+}) + c && \text{Definition 2.2.1} \\
&= ((a + b) + c){+}{+} && \text{Definition 2.2.1} \\
&= (a + (b + c)){+}{+} && \text{Inductive Hypothesis} \\
&= (a{+}{+}) + (b + c) && \text{Definition 2.2.1}
\end{aligned}$$

what closes the induction. □

Proposition 2.2.6. (Cancellation Law). *Let a, b, c be natural numbers such $a + b = a + c$. Then we have $b = c$.*

Proof. Let's induct on a. The base case, we need to prove $0 + b = 0 + c$ what leads to $b = c$ by Definition 2.2.1. For the step case, we need to prove if $(a{+}{+}) + b = (a{+}{+}) + c$, assuming $a + b = a + c$, then $b = c$.

$$
\begin{aligned}
(a{+}{+}) + b &= (a{+}{+}) + c && \\
(a + b){+}{+} &= (a + c){+}{+} && \text{Definition 2.2.1} \\
(a + b) &= (a + c) && \text{Axiom 2.4} \\
b &= c && \text{Inductive Hypothesis}
\end{aligned}
$$

□

Definition 2.2.7. (Positive natural numbers). *A natural number n is said to be positive iff it is not equal to* 0*.*

Proposition 2.2.8. *If a is positive and b is a natural number, then $a + b$ is positive.*

Proof. Let's induct on b. For the base case, $a + 0 = a$ by Definition 2.2.1. The result is positive by the propositional assumption. For the step case, we need to prove $a + (b{+}{+})$ is positive, assuming $a + b$ is positive. $a + (b{+}{+}) = (a + b){+}{+}$ by Lemma 2.2.3. Given Axiom 2.3, $(a + b){+}{+}$ is positive as well, what closes the induction. □

Corollary 2.2.9. *If a and b are natural numbers such that $a + b = 0$, then $a = 0$ and $b = 0$.*

Proof. Let's assume that $a \neq 0$ or $b \neq 0$. In either case, $a + b$ is a positive number by Proposition 2.2.8 what contradicts $a + b \overset{\lightning}{=} 0$. □

Lemma 2.2.10. *Let a be a positive number. Then there exists exactly one natural number b such that $b{+}{+} = a$.*

Proof. Suppose for sake of contradiction, there is another natural number c such that $c{+}{+} = a$. Then $b{+}{+} = c{+}{+}$, what makes $b = c$ by Axiom 2.4. □

Definition 2.2.11. (Ordering of the natural numbers). *Let n and m be natural numbers. We say that n is greater than or equal to m, and write $n \geq m$ or $m \leq n$, iff we have $n = m + a$ for some natural number a. We say that n is strictly greater than m, and write $n > m$ or $m < n$, iff $n \geq m$ and $n \neq m$.*

Lemma 2.2.11.a. *For any natural number n, $n{+}{+} > n$.*

Proof. For any natural number, we know that $n{+}{+} = n + 1$ from Corollary 2.2.3.a. Then using the fact that $n \neq n{+}{+}$ from Lemma 2.1.8.a, we can conclude $n{+}{+} > n$ by Definition 2.2.11. □

Proposition 2.2.12. (Basic properties of order for natural numbers). *Let a, b, c be natural numbers. Then*

Proposition 2.2.12.a. (Order is reflexive) $a \geq a$.

Proof. Any natural number a can be represented $a = a + 0$ using Lemma 2.2.2 what proves by Definition 2.2.11 that $a \geq a$. $\square$

Proposition 2.2.12.b. (Order is transitive) *If* $a \geq b$ *and* $b \geq c$*, then* $a \geq c$.

Proof. Given Definition 2.2.11 and the propositional assumption, b can be represented as $c+\tilde{c}$ and a can be represented as $b+\tilde{b}$ where $\tilde{b}$ and $\tilde{c}$ are natural numbers. Then combining both representations, $a = b+\tilde{b} = (c+\tilde{c})+\tilde{b} = c+(\tilde{c}+\tilde{b})$ using Proposition 2.2.5. $\tilde{c}+\tilde{b}$ is a natural number by Proposition 2.2.1.a what leads us to $a \geq c$ by Definition 2.2.11. $\square$

Proposition 2.2.12.c. (Order is antisymmetric) $a \geq b$ *and* $b \geq a$ *iff* $a = b$.
(Original) if $a \geq b$ *and* $b \geq a$*, then* $a = b$.

Proof. The forward pass: by the propositional assumption and Definition 2.2.11, $a = b + \tilde{b}$ and $b = a + \tilde{a}$. Combining both terms, $a = b + \tilde{a} = (a + \tilde{a}) + \tilde{a}$ what equals to $a + (\tilde{a} + \tilde{a})$ using Proposition 2.2.5. Recalling Cancellation Law from Proposition 2.2.6, $(\tilde{a}+\tilde{a})$ must be equal to 0 since $a = a+0$ by Definition 2.2.1. By Corollary 2.2.9, $\tilde{a} = 0$. Plugging it back to $b = a + \tilde{a} = a + 0 = a$.

The reverse pass: if $a = b$, then $a \geq b$ and $b \geq a$. If $a = b$, we can use the reflexivity of order from Proposition 2.2.12.a to transform $a \geq a$ and $b \geq b$ to $a \geq b$ and $b \geq a$ respectively. That closes the proof in both directions. $\square$

Proposition 2.2.12.d. (Addition preserves order) $a \geq b$ *if and only if* $a + c \geq b + c$.

Proof. Let's start with forward case: if $a \geq b$, then $a + c \geq b + c$. If $a \geq b$, then we can represent $a = b + \tilde{b}$ by Definition 2.2.11. Plugging the expression into $a + c$ results in $(b + \tilde{b}) + c$. Using Proposition 2.2.5 and Proposition 2.2.4, we can transform the expression into $(b + c) + \tilde{b}$ what proves $a + c \geq b + c$ by Definition 2.2.11.

Now backward: if $a + c \geq b + c$, then $a + c = b + c + \tilde{b}$ by Definition 2.2.11. Then using Cancellation Law from Proposition 2.2.6, $a = b + \tilde{b}$ which is $a \geq b$ by Definition 2.2.11. $\square$

Proposition 2.2.12.e. $a < b$ *if and only if* $b = a + (d{+}{+})$ *for any natural number* d.

Proof. For the forward case we need to prove that if $a < b$, then $b = a + d$ and $d \neq 0$. If $a < b$, then $b = a + d$ and $b \neq a$ by Definition 2.2.11. Let's assume that d is not a positive natural number (i.e. $d = 0$). Then $a + d = a + 0 = a \overset{\lightning}{=} b$ by the antecedent $b \neq a$.

For the backward pass: $b = a + (d{+}{+}) = (a{+}{+}) + d$ by Lemma 2.2.3 and Definition 2.2.1. If $a = b$, then $b \neq (a{+}{+})$ by Axiom 2.4. If $b \neq (a{+}{+})$ and $b = (a{+}{+}) + d$, then $b > (a{+}{+})$ by Definition 2.2.11. If $b > (a{+}{+})$, then $b > a$ by Lemma 2.2.11.a and Proposition 2.2.12.b. If $a \neq b$ and $b = a + (d{+}{+})$, then $b > a$ by Definition 2.2.11. □

Proposition 2.2.12.f. *$a < b$ if and only if $a{+}{+} \leq b$.*

Proof. For forward case, we need to prove if $a{+}{+} \leq b$, then $a < b$. Combining Definition 2.2.11 and associativity from Proposition 2.2.5, we can infer $b = (a{+}{+}) + d = a + (d{+}{+})$ where $d{+}{+}$ is a positive natural number by Axiom 2.3 and Definition 2.2.7. Thus, we can conclude that $b > a$ by Proposition 2.2.12.e.

For the backward: if $a{+}{+} \leq b$, then $b = (a{+}{+}) + d = a + (d{+}{+})$ by Proposition 2.2.4. Since $d{+}{+} \neq 0$ by Axiom 2.3, then $d{+}{+}$ is a positive number. If $d{+}{+}$ is a positive number, then $a < b$ by Proposition 2.2.12.e. □

Lemma 2.2.12.g. *$0 \leq b$ for all natural numbers.*

Proof. b can be represented as $b = 0 + b$ by Definition 2.2.1, what makes it $0 \leq b$ by Definition 2.2.11. □

Proposition 2.2.13. (Trichotomy of order for natural numbers). *Let a and b be natural numbers. Then exactly one of the following statements is true: $a < b, a = b, a > b$.*

Proof. Let's start that **no more than one** statement holds at a time. If $a = b$, then neither $a < b$ nor $a > b$ holds by Definition 2.2.11. If $a > b$ and $a < b$, then by Definition 2.2.11, $a \geq b$, $a \leq b$ and $a \neq b$. If $a \geq$b and $a \leq b$, then $a \overset{\lightning}{=} b$ using Proposition 2.2.12.c, what contradicts the antecedent. Thus, no more than one statement holds at a time.

Now let's prove that there is **at least one** statement holds inducting on a. For the base case, $0 \leq b$ by Lemma 2.2.12.g. For the step case, suppose that one of the trichotomy statements holds for a. We need to prove whether any holds for $a{+}{+}$. If $a < b$, then $a{+}{+} \leq b$ by Proposition 2.2.12.f. If $a = b$, then $a{+}{+} > b$ by Lemma 2.2.11.a. If $a > b$, then $a = b + (d{+}{+})$ by Proposition 2.2.12.e. Then we can take the successor $a{+}{+} = (b + (d{+}{+})){+}{+}$ by Axiom 2.4, what equals to $b + (d{+}{+}){+}{+}$ by Definition 2.2.1. $(d{+}{+}){+}{+}$ is a positive number by Definition 2.2.7 and Axiom 2.3. Thus, $a{+}{+} > b$ still holds by Proposition 2.2.12.e. That covers all possible cases and closes the induction. □

Proposition 2.2.14. (Strong principle of induction). *Let m_0 be a natural number, and let $P(m)$ be a property pertaining to an arbitrary natural number m. Suppose that for each $n \geq m_0$, we have the following implication: if $P(m)$ is true for all natural numbers $m_0 \leq m < n$, then $P(n)$ is also true (in particular, this means that $P(m_0)$ is true, since in this case the hypothesis is vacuous). Then we can conclude that $P(n)$ is true for all natural numbers $n \geq m_0$.*

Proof. We'll induct on n. For the base case m_0: using Definition 2.2.11, $P(m)$ is true for $m_0 \leq m \leq m_0$ and $m \neq m_0$. If $m_0 \leq m$ and $m_0 \geq m$, then $m_0 \overset{\lightning}{=} m$ by Proposition 2.2.12.c, what contradicts the fact that $m \neq m_0$. That means that there are no such m that satisfy $m_0 \leq m < m_0$. Thus, $P(m)$ is vacuously true for $m_0 \leq m < m_0$.

For the step case, suppose $P(m)$ for $m_0 \leq m < n \implies P(n)$, we need to prove $P(m)$ for $m_0 \leq m < n{+\!+} \implies P(n{+\!+})$. Since $P(m)$ holds for $m_0 \leq m < n$ and for $m = n$ (due to the given implication on $P(n)$), $P(m)$ holds for $m_0 \leq m \leq n$. The latter term is equivalent to $m_0 \leq m < n{+\!+}$ by Proposition 2.2.12.f. That means $P(m)$ is true for $m_0 \leq m < n{+\!+}$, and it means $P(n{+\!+})$ is true. That closes the induction. □

Proposition 2.2.15. (Principle of backwards induction). *Let n be a natural number, and let $P(m)$ be a property pertaining to the natural numbers such that whenever $P(m{+\!+})$ is true, then $P(m)$ is true. Suppose that $P(n)$ is true. Prove that $P(m)$ is true for all natural numbers $m \leq n$.*

Proof. Let's induct on n. If $n = 0$, then $m \leq 0$. Combining with Lemma 2.2.12.g, we can conclude that $m = 0$ by Proposition 2.2.12.c. Since $P(n)$ and $n = m$ are true, we proved $P(m)$ for $m \leq n$.

For the step case, suppose the following implication holds: $P(n) \implies P(m)$ is true for $m \leq n$. Now we need to prove $P(n{+\!+}) \implies P(m)$ is true for $m \leq n{+\!+}$. Let's assume that $P(n{+\!+})$ is true, then $P(n)$ is also true by the "backward" property. If $P(n)$ is true, then $P(m)$ is true for $m \leq n$. Since $P(m)$ is true for $m \leq n$ and for $m = n{+\!+}$ (by our assumption), then $P(m)$ is true for $m \leq n{+\!+}$. That proves the implication and closes the induction. □

Proposition 2.2.16. (Principle of induction starting from the base case n). *Let n be a natural number, and let $P(m)$ be a property pertaining to the natural numbers such that whenever $P(m)$ is true, $P(m{+\!+})$ is true. Show that if $P(n)$ is true, then $P(m)$ is true for all $m \geq n$.*

Proof. Since $m \geq n$, then we can represent $m = n + k$. So, let's induct on k. For the base case $k = 0$, $m = n$. Since $P(n)$ is true and $m = n$, then $P(m)$ is also true.

For the step case, we assume if $P(n)$ is true, then $P(n+k)$ is true for all $n+k \geq n$. Now we need to prove that the same holds for $P(n + (k{+}{+}))$. Given the fact that $m = n + k$, $n + (k{+}{+}) = (n + k){+}{+}$ by Definition 2.2.1, and the property assumption, if $P(m)$ is true, then $P(m{+}{+})$ is also true, we can conclude that $P(n + (k{+}{+}))$ is true. That closes the induction. $\square$

2.3 Multiplication

Definition 2.3.1. (Multiplication of natural numbers). *Let m be a natural number. To multiply zero to m, we define $0 \times m := 0$. Now suppose inductively that we have defined how to multiply n to m. Then we can multiply $n{+}{+}$ to m by defining $(n{+}{+}) \times m := (n \times m) + m$.*

Proposition 2.3.1.a. *Product of two natural numbers is a natural number.*

Proof. Let's induct on n. The base case would be $n := 0$. By definition 2.3.1, $0 \times m := 0$ which is a natural number. Now suppose $n \times m$ is a natural number. Then $(n{+}{+}) \times m := (n \times m) + m$ by Definition 2.3.1. Since $(n \times m)$ is a natural number by the inductive hypothesis and m is a natural number, the sum of two natural numbers is also a natural number by Proposition 2.2.1.a. That closes the induction. □

Lemma 2.3.1.b. $1 \times n = n$ *for all natural numbers.*

Proof.

$$
\begin{aligned}
1 \times n &= (0{+}{+})n && \text{By Definition 2.1.3} \\
&= (0 \times n) + n && \text{By Definition 2.3.1} \\
&= 0 + n && \text{By Definition 2.3.1} \\
&= n && \text{By Definition 2.2.1}
\end{aligned}
$$

□

Lemma 2.3.2.a. *For any natural number n, $n \times 0 = 0$.*

Proof. Let's induct on n. For the base case $n := 0$, $n \times 0 = 0 \times 0 := 0$ by Definition 2.3.1. Now for the step case, let's assume that $n \times 0 = 0$. Then

$$
\begin{aligned}
(n{+}{+}) \times 0 : &= (n \times 0) + 0 && \text{Definition 2.3.1} \\
&= 0 + 0 && \text{Inductive Hypothesis} \\
&= 0 && \text{Definition 2.2.1}
\end{aligned}
$$

what closes the induction. □

Lemma 2.3.2.b. *For any natural numbers n, m, $m \times (n{+}{+}) = (m \times n) + m$.*

Proof. Let's induct on m. For the base case,

$$\begin{aligned} m \times (n{+}{+}) &= 0 \times (n{+}{+}) \\ &= 0 \\ &= (0 \times n) + 0 \end{aligned}$$

For the step case, we assume that $m \times (n{+}{+}) = (m \times n) + m$. Now we need to prove that $(m{+}{+}) \times (n{+}{+}) = ((m{+}{+}) \times n) + (m{+}{+})$.

$$\begin{aligned} (m{+}{+}) \times (n{+}{+}) &= (m \times (n{+}{+})) + (n{+}{+}) && \text{Definition 2.3.1} \\ &= ((m \times n) + m) + (n{+}{+}) && \text{Inductive Hypothesis} \\ &= (m \times n) + (m + (n{+}{+})) && \text{Proposition 2.2.5} \\ &= (m \times n) + ((n{+}{+}) + m) && \text{Proposition 2.2.4} \\ &= (m \times n) + n + (m{+}{+}) && \text{Proposition 2.2.3} \\ &= ((m \times n) + n) + (m{+}{+}) && \text{Proposition 2.2.5} \\ &= ((m{+}{+}) \times n) + (m{+}{+}) && \text{Definition 2.3.1} \end{aligned}$$

that closes the induction. □

Lemma 2.3.2.c. (Multiplication is commutative). *Let n, m be natural numbers. Then $n \times m = m \times n$.*

Proof. Let's induct on n. For the base case $n := 0$,

$$\begin{aligned} n \times m &= 0 \times m \\ &:= 0 && \text{Definition 2.3.1} \\ &= m \times 0 && \text{Lemma 2.3.2.a} \end{aligned}$$

For the step case, let's assume that $n \times m = m \times n$. Then we need to prove that $(n{+}{+}) \times m = m \times (n{+}{+})$

$$\begin{aligned} (n{+}{+}) \times m &:= (n \times m) + m && \text{Definition 2.3.1} \\ &= (m \times n) + m && \text{Inductive Hypothesis} \\ &= m \times (n{+}{+}) && \text{Lemma 2.3.2.b} \end{aligned}$$

what closes the induction. □

Lemma 2.3.3. (Natural numbers have no zero divisors). *Let n, m be natural numbers. Then $n \times m = 0$ if and only if at least one of n, m is equal to zero. In particular, if n and m are both positive, then nm is also positive.*

Proof. We start with the left to right case. If $n \times m = 0$, then we need to prove that either n or m is zero. Let's assume than neither n nor m is zero (i.e. we can use $n{+}{+}$ and $m{+}{+}$ instead). Then $(n{+}{+}) \times (m{+}{+}) = (n \times (m{+}{+})) + m{+}{+}$ (or $(m{+}{+}) + (n{+}{+}) = (m \times (n{+}{+})) + (n{+}{+}))$ by Definition 2.3.1. We can conclude that in both cases, adding a natural number with a positive natural number will result in positive number by Proposition 2.2.8. That contradicts to $(n{+}{+}) \times (m{+}{+}) \overset{\lightning}{=} 0$. Thus, either n or m must be zero.

The right to left case: if either n or m is zero, then $n \times m = 0 \times m := 0$ by Definition 2.3.1 or $n \times m = n \times 0 = 0 \times n = 0$ by Lemma 2.3.2.a. What closes both directions.

If both, n and m, are positive natural numbers (i.e. $n{+}{+}$ and $m{+}{+}$), then $(n{+}{+}) \times (m{+}{+})$ is also positive as shown above. $\square$

Proposition 2.3.4. (Distributive law). *For any natural numbers a, b, c, we have $a(b+c) = ab + ac$ and $(b+c)a = ba + ca$.*

Proof. The case $a(b+c) = ab + ac$. Let's induct on a. The base case,

$$\begin{aligned} 0(b+c) &= 0b + 0c \\ 0 &= 0 + 0 && \text{Definition 2.3.1} \\ 0 &= 0 && \text{Definition 2.2.1} \end{aligned}$$

The step case, we assume that $a(b+c) = ab+ac$ to prove $a{+}{+}(b+c) = (a{+}{+})b+(a{+}{+})c$

$$\begin{aligned} a{+}{+}(b+c) &= a(b+c) + (b+c) && \text{Definition 2.3.1} \\ &= ab + ac + b + c && \text{Inductive Hypothesis} \\ &= ab + b + ac + c && \text{Propositions 2.2.4 and 2.2.5} \\ &= (a{+}{+})b + (a{+}{+})c && \text{Definition 2.3.1} \end{aligned}$$

what closes the induction for the former case. The case $(b+c)a = ba + ca$. Again, inducting on a. For the base case:

$$\begin{aligned} (b+c)0 &= b0 + c0 \\ 0 &= 0 + 0 && \text{Lemma 2.3.2.a} \\ 0 &= 0 && \text{Lemma 2.2.2} \end{aligned}$$

For the step case, assuming that $(b+c)a = ba + ca$, we need to prove $(b+c)(a{+}{+}) =$

$b(a{+}{+}) + c(a{+}{+})$

$$\begin{aligned}
(b+c)(a{+}{+}) &= (b+c)a + (b+c) && \text{Lemma 2.3.2.b} \\
&= ba + ca + b + c && \text{Inductive Hypothesis} \\
&= ba + b + ca + c && \text{Proposition 2.2.4 and 2.2.5} \\
&= b(a{+}{+}) + c(a{+}{+}) && \text{Lemma 2.3.2.b}
\end{aligned}$$

what closes the induction for the latter case. □

Proposition 2.3.5. (Multiplication is associative). *For any natural numbers a, b, c we have $(a \times b) \times c = a \times (b \times c)$.*

Proof. Let's induct on a. The base case:

$$\begin{aligned}
(0 \times b) \times c &= 0 \times (b \times c) && \\
0 \times c &= 0 && \text{By Definition 2.3.1 and Proposition 2.3.1.a} \\
0 &= 0 && \text{By Definition 2.3.1}
\end{aligned}$$

For the step case, we need to prove that $((a{+}{+}) \times b) \times c = (a{+}{+}) \times (b \times c)$, given $(a \times b) \times c = a \times (b \times c)$:

$$\begin{aligned}
((a{+}{+}) \times b) \times c &= ((a \times b) + b) \times c && \text{By Definition 2.3.1} \\
&= (a \times b) \times c + (b \times c) && \text{By Proposition 2.3.4} \\
&= a \times (b \times c) + (b \times c) && \text{By the Inductive Hypothesis} \\
&= (a{+}{+}) \times (b \times c) && \text{By Definition 2.3.1}
\end{aligned}$$

what closes the induction. □

Proposition 2.3.6. (Multiplication preserves the order). *If a, b are natural numbers such that $a < b$, and c is positive, then $ac < bc$.*

Proof. Since c is positive, we can denote it as $\bar{c}{+}{+}$. Let's induct on $\bar{c}$. The base case:

$$\begin{aligned}
a &< b && \\
0 + a &< 0 + b && \text{By Definition 2.2.1 and Proposition 2.2.12.d} \\
(a \times 0) + a &< (b \times 0) + b && \text{By Definition 2.3.1} \\
a(0{+}{+}) &< b(0{+}{+}) && \text{By Definition 2.3.1 and Lemma 2.3.2.c} \\
a(\bar{c}{+}{+}) &< b(\bar{c}{+}{+}) &&
\end{aligned}$$

For the step case, given $a(\bar{c}{++}) < b(\bar{c}{++})$, we need to prove that $a((\bar{c}{++}){++}) < b((\bar{c}{++}){++})$:

$$\begin{aligned}
a(\bar{c}{++}) &< b(\bar{c}{++}) && \text{By the Inductive Hypothesis} \\
a(\bar{c}{++}) + \bar{c}{++} &< b(\bar{c}{++}) + \bar{c}{++} && \text{By Proposition 2.2.12.d} \\
a((\bar{c}{++}){++}) &< b((\bar{c}{++}){++}) && \text{By Lemma 2.3.2.c and Definition2.3.1}
\end{aligned}$$

□

Corollary 2.3.7. (Cancellation law). *Let a, b, c be a natural numbers such that $ac = bc$ and $c \neq 0$. Then $a = b$.*

Proof. Let's apply the contrapositive property of implication: if $a \neq b$ or, in other words, $a > b$ exclusively or $a < b$, then $ac \neq bc$ or $c = 0$. If $c \neq 0$ and $a > b$, then $ac \lessgtr bc$ by Proposition 2.3.6, which means $ac \neq bc$ by Proposition 2.2.13. Since Proposition 2.2.13 requires $c = 0$, we need to consider $c = 0$. If $c = 0$, then the consequent is trivially satisfied. □

Proposition 2.3.8. (Euclidean algorithm). *Let n be a natural number, and let q be a positive number. Then there exist natural numbers m, r such that $0 \leq r < q$ and $n = mq + r$.*

Proof. Now, we fix q and induct on n. For the base case $n = 0$, we have $0 = mq + r$. To satisfy the statement, both terms of RHS must be equal to zero. Indeed, there exist $m = 0$ and $r = 0$ that satisfy the statement and do not violate the condition $0 \leq r < q$. For the step case, we assume that the implication holds for n, and we need to prove that it holds for $n{++}$.

$$\begin{aligned}
n{++} &= (mq + r){++} && \text{Inductive Hypothesis} \\
&= mq + (r{++}) && \text{By Proposition 2.2.4 and Definition 2.2.1}
\end{aligned}$$

if $r{++} < q$, then $0 \leq r < r{++} < q$ by Lemma 2.2.11.a and Proposition 2.2.12.b. If $r{++} = q$, then $mq + (r{++}) = mq + q = (m{++})q + 0 = \bar{m}q + \bar{r}$ by Definition 2.3.1. The case $r{++} > q$ is not relevant due to Proposition 2.2.12.f. □

Definition 2.3.9. (Exponentiation for natural numbers). *Let m be a natural number. To raise m to the power 0, we define $m^0 := 1$. Now suppose recursively that m^n has been defined for some natural number n, then we define $m^{n++} := m^n \times m$.*

Proposition 2.3.10. $(a + b)^2 = (a^2 + 2ab + b^2)$ *for all natural numbers a, b.*

Proof.

$$
\begin{aligned}
(a+b)^2 &= (a+b)^{1{+}{+}} \\
&= (a+b)\times(a+b) && \text{By Definition 2.3.9} \\
&= (a+b)a + (a+b)b && \text{By Proposition 2.3.4} \\
&= aa + ab + ab + bb && \text{By Proposition 2.3.4} \\
&= a^2 + (1\times ab) + ab + b^2 && \text{By Definition 2.3.9 and Proposition 2.3.1.b} \\
&= a^2 + (1{+}{+})ab + b^2 && \text{By Definition 2.3.1} \\
&= a^2 + 2ab + b^2 && \text{By Definition 2.1.3}
\end{aligned}
$$

□

Chapter 3

Set Theory

3.1 Fundamentals

Axiom 3.1. (Sets are objects.) *If A is a set, then A is also an object. In particular, given two sets A and B, it is meaningful to ask whether A is also an element of B.*

Axiom 3.2. (Equality of sets). *Two sets A and B are equal, $A = B$, iff every element of A is an element of B and vice versa. To put it another way, $A = B$ if and only if every element x of A belongs also to B, and every element y of B belongs also to A.*

Axiom 3.3. (Empty set). *There exists a set $\emptyset$ or $\{\}$, known as the empty set, which contains no elements, i.e. for every object x we have $x \notin \emptyset$.*

Proposition 3.1.1. *There is exactly one empty set $\emptyset$.*

Proof. Let us assume that $\emptyset'$ is also an empty set. Based on Axiom 3.3, since no elements are in both sets, then all elements (in particular, the lack of them), that are in $\emptyset$, are also in $\emptyset'$ and vice versa. That makes $\emptyset' = \emptyset$. □

Lemma 3.1.5. (Single choice). *Let A be a non-empty set. Then there exists an object x such that $x \in A$.*

Proof. Let us prove by contradiction. If A is a non-empty set, then no x exists such that $x \in A$. Or in other words, for every object x we have $x \notin A$. But this is the empty set by Axiom 3.3. □

Axiom 3.4. (Singleton sets and pair sets). *If a is an object, then there exists a set $\{a\}$ whose only element is a, i.e., for every object y, we have $y \in \{a\}$ if and only if $y = a$; we refer to a as the singleton set whose element is a. Furthermore, if a and b are objects, then there exists a set $\{a, b\}$ whose only elements are a and b; i.e., for every object y, we have $y \in \{a, b\}$ if and only if $y = a$ or $y = b$; we refer to this set as the pair set formed by a and b.*

Proposition 3.1.6. *There is only one singleton set for each object a.*

Proof. Let's assume that there's another singleton set a' for a different from a which is a singleton set of a. Since only $a \in a'$ by Axiom 3.4, then for all $x \in a$ it is also $x \in a'$. What makes $a = a'$ by Axiom 3.2 what contradicts our initial assumption. □

Proposition 3.1.7. *Given two objects a and b, there is only one pair set formed by a and b.*

Appendix A

The basics of mathematical logic

A.1 Mathematical statements

Exercise A.1.1. *What is the negation of the statement "either X is true or Y is true, but not both"?*

Solution. X and Y are false or X and Y are true. □

Exercise A.1.2. *What is the negation of the statement "X is true if and only if Y is true"?*

Solution. X is false iff Y is false □

Exercise A.1.3. *Suppose that you have shown that whenever X is true, then Y is true, and whenever X is false, then Y is false. Have you demonstrated that X and Y are logically equivalent? Explain.*

Solution. A well-defined statement can be either true or false only (all possible values). Now, we consider the assumptions: (1) if X is false, then Y is false (Y is never true in that case), and (2) if X is true, then Y is true (Y is never false here). Thus, both statements coincide in all possible values what makes them "equivalent". □

Exercise A.1.4. *Suppose that you have shown that whenever X is true, then Y is true, and whenever Y is false, then X is false. Have you demonstrated that $X \Leftrightarrow Y$? Explain.*

Solution. No. If X is true, then Y is true: we exclude only the case X is true and Y is false. If Y is false, then X is false: we exclude the case X is true and Y is false. The both cases represent the same statement. As the result, the case X is False and Y is True is missing. Thus, X and Y are not being proven to be logically equivalent. □

Exercise A.1.5. *Suppose that you have shown that X is true iff Y is true, and you know that Y is true iff Z is true. Is this enough to show that X, Y, Z are logically equivalent? Explain.*

Solution. Yes. Let's construct a logical table of all possible cases:

X	Y	Z	Possible?	If not, why?
False	False	False	Yes	
False	False	True	No	$Y \Leftrightarrow Z$
False	True	False	No	$X \Leftrightarrow Y$ or $Y \Leftrightarrow X$
False	True	True	No	$X \Leftrightarrow Y$
True	False	False	No	$X \Leftrightarrow Y$
True	False	True	No	$X \Leftrightarrow Y$ or $Y \Leftrightarrow X$
True	True	False	No	$Y \Leftrightarrow Z$
True	True	True	Yes	

What results X, Y, Z to be logically equivalent by definition. □

Exercise A.1.6. *Suppose that you have shown that whenever X is true, then Y is true; that if Y is true, then Z is true; and if Z is true, then X is true. Is this enough to show that X, Y, Z are logically equivalent? Explain.*

Solution. Yes. Let's construct a logical table of all possible cases:

X	Y	Z	Possible?	If not, why?
False	False	False	Yes	
False	False	True	No	$Y \implies Z$
False	True	False	No	$X \implies Y$
False	True	True	No	$Z \implies X$
True	False	False	No	$X \implies Y$
True	False	True	No	$X \implies Y$
True	True	False	No	$Y \implies Z$
True	True	True	Yes	

What results X, Y, Z to be logically equivalent by definition. □

A.2 Implication

The main ideas in the chapter are the following:

- Vacuous hypothesis is a hypothesis where an antecedent is false. Since false implies false/true is always true, it does not provide any new information.

- We can also draw the following conclusions:

$$\begin{aligned} X \implies Y &\iff \neg Y \implies \neg X && \text{Contrapositive} \\ X \implies Y &\not\Longleftrightarrow Y \implies X && \text{Converse} \\ X \implies Y &\not\Longleftrightarrow \neg X \implies \neg Y && \text{Inverse} \end{aligned}$$

- Although, vacuous hypothesis can be viewed as "useless", it can still be used for a proof by contradiction using Contrapositive property, for instance. For that case, we need to have a consequent of the original implication to be known to be false.

A.3 The structure of proofs

No exercise in this section.

A.4 Variables and Quantifiers

No exercise in this section.

A.5 Nested Quantifiers

Exercise A.5.1(a). *For every positive number x, and every positive number y, we have $y^2 = x$.*

Solution. For any real number x and y, $y^2 = x$. False. Let's prove by counter example. If $x = 1, y = 2$, then $2^2 \overset{\lightning}{=} 1$. A game metaphor would be "Your opponent picks real numbers x, y. If you can show that $y^2 = x$ in every game, you win". $\square$

Exercise A.5.1(b). *There exists a positive number x such that for every positive number y, we have $y^2 = x$.*

Solution. One can pick a positive number x s.t. $y^2 = x$ for any positive y. False. Let's assume that such x exists and we pick two positive numbers y_1 and y_2, for instance, 2 and 3 respectively. Then we have $y_1^2 = 4 = x$ and $y_2^2 = 9 = x$. If $4 = x$ and $9 = x$, then $4 \overset{\lightning}{=} 9$. A game metaphor would be "You pick a positive number x. Then an opponent pick a positive number y. If you can show that $y^2 = x$ in every game, you win". $\square$

Exercise A.5.1(c). *There exists a positive number x, and there exists a positive number y, such that $y^2 = x$.*

Solution. One can pick two positive numbers x and y, s.t. $y^2 = x$. True. We just need to show an example. Let's assume that $x = 1$ and $y = 1$. Then $1^1 = 1 = 1$. A game metaphor would be "You pick two positive numbers x and y. If you can show that $y^2 = x$ in every game, you win". □

Exercise A.5.1(d). *For every positive number y, there exists a positive number x such that $y^2 = x$.*

Solution. For any a positive number y, one can pick a positive number x s.t. $y^2 = x$ is true. True. *(Informal Proof).* Since we know the value y, we can derive x as a function of $y : x(y) = y^2$. Since the function's domain is $(0, \infty)$ for $y \in (0, \infty)$, then for any positive y we can find $x(y) = y^2$. As a game metaphor "An opponent picks a positive number y. You need to select a positive number x. If you can show every time that $y^2 = x$, then you win". □

Exercise A.5.1(e). *There exists a positive number y such that for every positive number x, we have $y^2 = x$.*

Solution. One can pick a positive number y such that for any positive number x, the statement $y^2 = x$ is true. False. Let's assume that such y exists. Then we pick positive numbers x_1 and x_2 s.t. $x_1 \neq x_2$. For instance, 1 and 2 respectively. If $x_1 = 1, x_2 = 4$, then $y^2 = 1$ and $y^2 = 4$. If $y^2 = 1$ and $y^2 = 4$, then $1 \overset{\lightning}{=} 4$. A game metaphor in this case would be "You pick a positive number y, then an opponent picks a positive number x. If you can show that $y^2 = x$ in every game, you win". □

Appendix B

The decimal system